Roll No.: .....

END SEMESTER EXAMINATION DECEMBER 2025

Name of the Program: M.Tech | Semester: I

Name of the Course: Cloud Computing | Course Code: MCS 125

Time: 3 Hours | **Maximum Marks: 100**

Note:

i. **All questions are compulsory.**

ii. Answer any two sub questions among a, b & c in each main questions

iii. Total marks in each main question are twenty.

iv. Each questions carries 20 marks

Q1	(20Marks)	
(a)	Interpret the NIST reference architecture and evaluate its relevance in establishing interoperability standards. A small business is evaluating whether to move its applications to the cloud. Using NIST's essential characteristics explain how these will benefit the business. Support your answer with typical diagram.	CO1 CO1
(b)	Discuss how distributed computing forms the foundation of cloud computing services.	CO1
(c)	Classify various deployment models of cloud computing with neat diagrams. ShopEase migrated its e-commerce operations to the cloud. Based on this case, discuss the advantages and disadvantages of cloud computing for businesses	
Q2		
(a)	Illustrate how virtualization improves data-centre efficiency and assess issues related to multitenancy. Additionally discuss the Types of Data centre with their use cases.	CO2
(b)	Company X is evaluating *Type 1 and Type 2 hypervisors* for production and development use. Discuss hypervisor with examples. Compare and contrast between the two types of hypervisors, their advantages and disadvantages with typical block diagrams .	CO2 CO2
(c)	Discuss *KVM and Xen* and differentiate their architectural approaches, use cases advantages and limitations.	
Q3		
(a)	Explain the architecture of *Mobile Cloud Computing* and illustrate how its mobile devices, cloud infrastructure, and network interfaces interact to deliver seamless computing and storage services.	CO4
(b)	Compare and contrast the architectures and operational principles of *Edge Computing* and *Fog Computing*, and evaluate how each model addresses latency, scalability, and data-processing challenges in cloud environments.	CO3
(c)	Write short notes on the following in the context of Cloud Computing (*any two*) 1. Vendor lock-in 2. Interoperability 3. Metering Issues and Report generation	
Q4		
(a)	Analyse the importance of *Service Level Agreements (SLA)* in cloud computing and discuss their impact on service reliability and customer trust. Discuss the *SLA lifecycle* with a typical diagram.	CO5 CO5 CO4

(b)	Discuss how load balancing and scaling work together in cloud computing to manage varying workloads and maintain system performance.	
(c)	Analyze the concept of *dynamic provisioning* in cloud computing and explain how effective resource management ensures optimal performance and cost efficiency.	
Q5		
(a)	Discuss cloud migration and explain why it is needed in cloud environments.	CO5
(b)	A startup must select between AWS, Google Cloud, Azure, and IBM Cloud for global deployment. Compare these providers in terms of *services, scalability, compliance, pricing and enterprise support features.*	CO6
(c)	Describe the architecture of *Hadoop* with typical diagram. Illustrate the *MapReduce* programming model.	CO6